

Tutorial 1: Review Object Oriented Programming concepts

(*) You should follow instructor's guideline to complete tutorial in correct format. Write your answer in section "Your answer goes here". In this section, font set to "Times new roman", with normal font, code example MUST BE in Courier New.

Part 2: Practice exercises

From this part onwards, you are required to program based on requirements below. Since it's the first tutorial, work in group of 3 or less is required. You are free to choose your preferred IDE to finish tasks.

Req 1: a. Create a class called Car. The Car class has the following fields and methods.

- int speed;
- Double regularPrice;
- String color;
- String carName → not in constructor, but in setter method.
- Double getSalePrice(). Assume if color is "Black", then sale price is equal to "Regular price * 1.5", else equal "Regular price * 1.2"

Req 2: Create instance of class Car, named "Mondeo" with following value, using constructor to initialize object of class

Speed = 80

regularPrice = 5000

color = red

Print to screen car information, such as "**Mondeo** car whose speed is **80**, color **Red** with regular price = **5000** and sale price = **6000**"

Req 3: Create instance of Car, named "Camry" with following value, using constructor to initialize object of class

Speed = 90

regularPrice = 6000

color = black

Print to screen car information, such as “**Camry** car whose speed is **80**, color **Black** with regular price = **6000** and sale price = **9000**”

Req 4: This is new requirement, it's not related to previous exercise.

Create class **Student** and **Course** including these attributes, write getters and setters methods if necessary for these attributes of each class.

`Student:`

- + `String studentId`
- + `String studentName`

`Course:`

- + `String courseCode`
- + `String courseName`

And create `Enrolment` class, which contain information of student and courses with corresponding marks. Attributes of `Enrolment` class include

- + `Student student`
- + `Course course`
- + `int attendanceMark`
- + `float midtermMark`
- + `float finalMark`

Initiate 3 students and 3 courses with attribute values of your choice, and assign Student to courses with attendance mark, midterm and final mark of your choice. Finally print out report as follow

"Student <name> whose id is <studentId>, enrol in Course <name> got attendance mark = <attendance mark>, midterm = <midterm> and final = <final>. Overall is <overall>".

(*) Note: Overall = (attendance * 1 + midterm * 3 + final * 6)/10

Answer

1+2+3 Class Car

```
public class Car
{
    private int speed;
    private double regularPrice;
    private String color;
    private String carName;

    public Car(int speed, double regularPrice, String color)
    {
        this.speed = speed;
        this.regularPrice = regularPrice;
        this.color = color;
    }

    public void setCarName(String name)
    {
        this.carName = name;
    }

    public double getSalePrice()
    {
        if (this.color == "black") {
            return this.regularPrice * 1.5;
        }
        return this.regularPrice * 1.2;
    }

    public String toString()
    {
        return this.carName + " car whose speed is " + this.speed + ", color " + this.color + " with regular price = " + this.regularPrice + " and sale price = " + this.getSalePrice();
    }
}
```

Main

```

public class MyClass
{
    public static void main(String args[])
    {
        Car car = new Car(80,5000,"red");
        car.setCarName("Mondeo");
        System.out.println(car.toString());

        Car car2 = new Car(90,6000,"black");
        car2.setCarName("Toyota");
        System.out.println(car2.toString());
    }
}

```

Req 4

```

public class Course {
    private String courseCode;
    private String courseName;

    public Course(String courseCode, String courseName)
    {
        this.courseCode = courseCode;
        this.courseName = courseName;
    }

    public String getName()
    {
        return this.courseName;
    }
}

```

```
public class Student {  
    private String studentId;  
    private String studentName;  
  
    public Student(String studentId, String studentName)  
    {  
        this.studentId = studentId;  
        this.studentName = studentName;  
    }  
  
    public String getName()  
    {  
        return this.studentName;  
    }  
  
    public String getStudentId()  
    {  
        return this.studentId;  
    }  
}
```

```
public class Enrolment
{
    private Course course;
    private Student student;
    private int attendanceMark;
    private double midtermMark;
    private double examMark;

    public Enrolment(Student student, Course course)
    {
        this.course = course;
        this.student = student;
    }

    public void setAttendanceMark(int attendanceMark)
    {
        this.attendanceMark = attendanceMark;
    }

    public void setMidtermMark(double midtermMark)
    {
        this.midtermMark = midtermMark;
    }

    public void setExamMark(double examMark)
    {
        this.examMark = examMark;
    }

    public double calculateOverall()
    {
        return (this.attendanceMark + this.midtermMark * 3 + this.examMark * 6)/10;
    }

    public String toString()
    {
        Student s = this.student;
        Course c = this.course;
        return "Student " + s.getName() + " whose id is " + s.getId() + ", enroll course " + c.getName() +
            " got attendance mark = " + this.attendanceMark +
            ", midterm = " + this.midtermMark +
            " and final = " + this.examMark +
            ". Overall is " + this.calculateOverall();
    }
}
```

```
public class MyClass {  
    public static void main(String args[]) {  
  
        Student s = new Student("0901040005", "camnh");  
        Course c = new Course("IPG", "Introduction to programming");  
  
        Enrolment e = new Enrolment(s, c);  
        e.setAttendanceMark(4);  
        e.setMidtermMark(7.5);  
        e.setExamMark(8);  
  
        System.out.println(e.toString());  
    }  
}
```

Student camnh whose id is 0901040005, enroll course Introduction to programming got attendance mark = 4, midterm = 7.5 and final = 8.0. Overall is 7.45